Module 16: Event Handling

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Prerequisites:

1. To complete this lab you will be using the VM provided to you.
    a. Login: builtin\AXLocalAdmin
    b. Password: <Available in LCS>
2. Open Visual Studio as administrator
    a. Right Click Visual Studio icon, Run as administrator
3. Exercises
    a. Model Creation & project Creation
    b. Label File Creation
    c. Base Enumeration Creation
    d. Extended Data Type Creation
    e. Table Creation
    f. Form Creation

Scenario

For Asset Insurance System scenario, there are few fields added in the standard tables, as following

1. Insurance Vendor Group in the Vendor Parameter Table
2. Premium Calculation Type in the Vendor Master table
3. Premium payment cycle in the Asset master table

Based on these fields there are few validations on the existing forms.

- If the Insurance Vendor is changed in the AssetTable, Policy Amount should become 0

Event handling is being used to address these validations

Practice: AssetTable table modifiedField post Event

1. In the Solution Explorer, right click Asset Insurance System project
2. Click Add > New Item
3. Under Operations Artifacts, Click Code
4. Select Class
5. Name: aisAssetTableEventHandler
6. Click Add
7. Open Application Explorer from the View menu in VS
8. Find AssetTable under table node and open
9. Right Click AssetTable > Methods > modifiedField and click on Copy Event Handler method > Post-event handler
10. Open aisAssetTableEventHandler class and Paste
11. You will find a method signature is pasted in the class as follows

```
/// <summary>
///
/// </summary>
/// <param name="args"></param>
[PostHandlerFor(tableStr(AssetTable), tableMethodStr(AssetTable, modifiedField))]
public static void AssetTable_Post_modifiedField(XppPrePostArgs args)
```

12. Add following code in the method

```
[PostHandlerFor(tableStr(AssetTable), tableMethodStr(AssetTable, modifiedField))]
public static void AssetTable_Post_modifiedField(XppPrePostArgs args)
{
    AssetTable assetTable;
    int fldNum;

    fldNum = args.getArgNum(1);
    assetTable = args.getThis();
    switch (fldNum)
    {
        case Fieldnum(assetTable, InsuranceVendor):
            {
                assetTable.PolicyAmount = 0;
                break;
            }
    }
}
```

Practice: AssetTable Form Datasource Activated event

1. Open Application Explorer from the View menu in VS
2. Find AssetTable under form node and open
3. Right Click AssetTable > datasource > AssetTable > Events > OnActivated and click on Copy Event Handler method
4. Open aisAssetTableEventHandler class and Paste
5. You will find a method signature is pasted in the class as follows

```
/// <summary>
///
/// </summary>
/// <param name="sender"></param>
/// <param name="e"></param>
[FormDataSourceEventHandler(formDataSourceStr(AssetTable, AssetTable), FormDataSourceEventType::Activated)]
public static void AssetTable_OnActivated(FormDataSource sender, FormDataSourceEventArgs e)
```

6. Add following code in the method

```
[FormDataSourceEventHandler(formDataSourceStr(AssetTable, AssetTable), FormDataSourceEventType::Activated)]
public static void AssetTable_OnActivated(FormDataSource sender, FormDataSourceEventArgs e)
{
    AssetTable        assetTable = sender.cursor();
    FormRun         element = sender.formRun();
    FormControl     insCalcPolicy = element.design(0).controlName("aisInsurance");

    if(assetTable.InsuranceVendor)
        insCalcPolicy.enabled(true);
    else
        insCalcPolicy.enabled(false);
}
```

Practice: AssetTable Form Insurance Vendor lookup event

7. Open Application Explorer from the View menu in VS

8. Find AssetTable under form node and open
9. Right Click AssetTable > design > Tab (Tab) > TabPageDetails (tab page) > TabHeader (Tab) > InsuranceTab (Tab page) > Insurance (Group) > Insurance_InsuranceVendor > Events > OnLookup and click on Copy Event Handler method
10. Open aisAssetTableEventHandler class and Paste
11. You will find a method signature is pasted in the class as follows

```
/// <summary>
///
/// </summary>
/// <param name="sender"></param>
/// <param name="e"></param>
[FormControlEventHandler(formControlStr(AssetTable, Insurance_InsuranceVendor), FormControlEventType::Lookup)]
public static void Insurance_InsuranceVendor_OnLookup(FormControl sender, FormControlEventArgs e)
```

12. Add following code in the method

```
[FormControlEventHandler(formControlStr(AssetTable, Insurance_InsuranceVendor), FormControlEventType::Lookup)]
public static void Insurance_InsuranceVendor_OnLookup(FormControl sender, FormControlEventArgs e)
{
    Query query = new Query();
    QueryBuildDataSource qbds_VendTable;
    SysTableLookup sysTableLookup;
    QueryBuildRange qbr;

    sysTableLookup = SysTableLookup::newParameters(tablenum(VendTable), sender);
    sysTableLookup.addLookupfield(fieldnum(VendTable, AccountNum), true);

    qbds_VendTable = query.addDataSource(tableNum(VendTable));

    qbr = qbds_VendTable.addRange(fieldNum(VendTable, VendGroup));
    qbr.value(VendParameters::find().aisInsuranceVendorGroup);

    sysTableLookup.parmQuery(query);
    sysTableLookup.performFormLookup();
}
```

Practice: VendTable Form Datasource Activated event

13. In the Solution Explorer, right click Asset Insurance System project
14. Click Add > New Item
15. Under Operations Artifacts, Click Code
16. Select Class
17. Name: aisVendTableEventHandler
18. Click Add
19. Open Application Explorer from the View menu in VS
20. Find VendTable under form node and open
21. Right Click VendTable > datasource > VendTable > Events > OnActivated and click on Copy Event Handler method
22. Open aisVendTableEventHandler class and Paste
23. You will find a method signature is pasted in the class as follows

```
/// <summary>
///
/// </summary>
/// <param name="sender"></param>
/// <param name="e"></param>
[FormDataSourceEventHandler(formDataSourceStr(VendTable, VendTable), FormDataSourceEventType::Activated)]
public static void VendTable_OnActivated(FormDataSource sender, FormDataSourceEventArgs e)
```

24. Add following code in the method

```
[FormDataSourceEventHandler(formDataSourceStr(VendTable, VendTable), FormDataSourceEventType::Activated)]
public static void VendTable_OnActivated(FormDataSource sender, FormDataSourceEventArgs e)
{
    FormRun         element = sender.formRun();
    FormControl     vendGroup = element.design(0).controlName("Posting_VendGroup");
    FormControl     insCalc = element.design(0).controlName("aisAssetInsurance_aisInsuranceCalc");

    if(vendGroup.valueStr() == VendParameters::find().aisInsuranceVendorGroup)
        insCalc.enabled(true);
    else
        insCalc.enabled(false);
}
```

Practice: VendTable Form Vendor Group modified event

25. Open Application Explorer from the View menu in VS
26. Find VendTable under form node and open
27. Right Click VendTable > design > MainTab (Tab) > TabPageDetails (Tab page) > Tab (Tab) > TabGeneral (Tab page) > UpperGroup (Group) > Posting (Group) > Posting_VendGroup > Events > OnModified and click on Copy Event Handler method
28. Open aisVendorTableEventHandler class and Paste
29. You will find a method signature is pasted in the class as follows

```
/// <summary>
///
/// </summary>
/// <param name="sender"></param>
/// <param name="e"></param>
[FormControlEventHandler(formControlStr(VendTable, Posting_VendGroup), FormControlEventType::Modified)]
public static void Posting_VendGroup_OnModified(FormControl sender, FormControlEventArgs e)
```

30. Add following code in the method

```
[FormControlEventHandler(formControlStr(VendTable, Posting_VendGroup), FormControlEventType::Modified)]
public static void Posting_VendGroup_OnModified(FormControl sender, FormControlEventArgs e)
{
    FormRun         element = sender.formRun();
    FormControl     insCalc = element.design(0).controlName("aisAssetInsurance_aisInsuranceCalc");

    if(sender.valueStr() == VendParameters::find().aisInsuranceVendorGroup)
        insCalc.enabled(true);
    else
        insCalc.enabled(false);
}
```

Try yourself

Add a post event handler in the modified field method of Vendtable, to set the value of Premium Calculation type as None, in case the Vendor Group selected is not for insurance vendor group

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